

Graphs 5

Consolidation

6. $x^2 + y^2 = 10$

For the tangents that have the equation $y = 3x + k$,

$$x^2 + (3x + k)^2 = 10$$

$$x^2 + 9x^2 + 6kx + k^2 = 10$$

$$10x^2 + 6kx + (k^2 - 10) = 0$$

Then, the discriminant $b^2 - 4ac < 0$ when the line does not intersect.,

$$(6k)^2 - 4 \times 10 \times (k^2 - 10) < 0$$

$$36k^2 - 40k^2 + 400 < 0$$

$$-k^2 + 100 < 0$$

$$(k + 10)(10 - k) < 0$$

So $k > 10$ or $k < -10$.

Exam Questions

1. A: (3, 0). B: (9, 8).

a. Then the center of the circle is at (6, 4).

b. Gradient of line AB is $\frac{\Delta y}{\Delta x} = \frac{8}{6} = \frac{4}{3}$. So the tangent at B will have the gradient $-\frac{3}{4}$.

Then,

$$y - y_1 = m(x - x_1)$$

$$y - 8 = -\frac{3}{4}(x - 9)$$

$$4y - 32 = -3x + 27$$

$$4y + 3x - 59 = 0$$

2. a. $y = 2x$

$$\rightarrow x^2 + (2x)^2 + 2x - 16(2x) + 56 = 0$$

$$5x^2 - 30x + 56 = 0$$

If the line does not meet the circle, the discriminant $b^2 - 4ac < 0$.

$$\begin{aligned} (-30)^2 - 4 \times 5 \times 56 &= 900 - 1120 \\ &= -220 < 0 \end{aligned}$$

So the line does not meet the circle.

b. If the line is perpendicular to $y = 2x$, the gradient is $-\frac{1}{2}$.

First, let's rearrange the equation:

$$x^2 + 2x + y^2 - 16y + 56 = 0$$

$$(x + 1)^2 + (y - 8)^2 - 9 = 0$$

So the center of the circle has coordinates $(-1, 8)$.

Then,

$$y - y_1 = m(x - x_1)$$

$$y - 8 = -\frac{1}{2}(x + 1)$$

$$2y - 16 = -x - 1$$

$$x + 2y - 15 = 0$$

3. a. $(x - a)^2 + (y + a)^2 - 2a^2 = 0$

b. If $(1, -5)$ lies on the circle, $(1 - a)^2 + (-5 + a)^2 - 2a^2 = 0$.

$$1 - 2a + a^2 + 25 - 10a + a^2 - 2a^2 = 0$$

$$-12a + 26 = 0$$

$$a = \frac{13}{6}$$

So the center of the circle is at $(\frac{13}{6}, -\frac{13}{6})$, and the radius is $\sqrt{2(\frac{13}{6})^2} = \frac{13}{3\sqrt{2}}$.

So the area of the circle is $2\pi \times \frac{169}{36} = \frac{169}{18}\pi$